FULL-TIME PROFESSOR OF PRACTICE • INTERACTION & ENTERPRISE SYSTEMS

# Bridging Enterprise HCI, Agentic AI & Design Systems with Studio Pedagogy

Body of Work, Practice-Led Curriculum Proposals, and 3-Year Departmental Growth Plan aligned with IIT Hyderabad's Fractal Academics & Agenda 40.6.1 Framework.

| | | | |
|---|---|---|---|
| **18+ Yrs** | **25+** | **50+** | **1.5 Yrs** |
| Industry Practice | Products Shipped | Designers Mentored | Engg. Faculty Roots |



**Balaajee V.R.**

**Senior UI/UX Design Manager (Ex-IBM)**

B.E. in Information Technology • MS University

IBM Software Labs • Zebra Technologies
ZF Wabco • Ericsson • Infosys
Former Lecturer (Anna Univ. Affiliated)

# A Full-Circle Return to Teaching — Plugging a High-Demand Industry Gap

I began my career in the engineering classroom before spending 18+ years building complex global software. Returning full-time to IIT Hyderabad lets me bring live enterprise practice directly into B.Des, M.Des, and PhD studios.

1. FULL-CIRCLE ACADEMIC FOUNDATION

## From Engineering Lecturer to Global Design Leader

- **Classroom Roots (2004–2006):** Served 1.5 years as Lecturer (IT) & Teaching Faculty at an Anna University affiliated engineering college — already grounded in syllabus design, grading rubrics, and student mentoring.
- **Engineering + Design Fluency:** Hold a **B.E. in Information Technology** plus 4+ years in software engineering at Infosys before 14+ years leading global UX — allowing effortless collaboration across Design and CSE/AI departments.
- **Mentorship at Scale:** Hired, coached, and reviewed portfolios for **50+ product designers**, most recently managing a 14-person multidisciplinary UX studio at IBM Software Labs.
- **Full-Time Kandi Commitment:** Ready to relocate and dedicate 100% of my professional focus on-campus at IIT Hyderabad.

2. COMPLEMENTING DOD IITH'S STRENGTHS

## Adding Enterprise Systems, Agentic AI & DesignOps Depth

DoD IITH already excels in Visual Communication, Film & Animation, Cultural Heritage, Ergonomics, and Industrial Design. My practice adds complementary depth where top tech employers hire heavily:

**Agentic & Human-in-the-Loop AI**
Autonomy calibration, reasoning traces, error recovery & AI quality evals.

**Design Systems & Token CI/CD**
Multi-brand Figma-to-React/Web-Component pipelines at 45-squad scale.

**Safety-Critical IoT & Mobility HMI**
Sub-second truck cockpit glance budgets & rugged industrial handhelds.

**Enterprise SaaS & GCC Network**
Direct bridges to Hyderabad, Bengaluru & global product design orgs.

# 18+ Years Across Global R&D Labs — Practicing Studio Critique Every Week

Exceeding IITH's 15-year practitioner benchmark across cloud software, cybersecurity, telecom, industrial IoT, automotive cockpits, and healthcare UX research.

**IBM Software Labs**
Sr. Design Manager • 2024–26
Directed 14-designer studio; Watsonx Agentic AI UX & Carbon token governance.

**Zebra Technologies**
Product Design Mgr • 2021–24
38,000+ rugged IoT handhelds, VisibilityIQ fleet UX & Lean UX sprints.

**ZF Wabco**
UI/UX Sr. Specialist • 2020–21
Commercial truck cockpit HMI & telematics under <1.2s safety glance budgets.

**Ericsson Global**
UI/UX Designer • 2019–20
45-country Telecom CPQ & Design System; 1st Place Global Hack-fest.

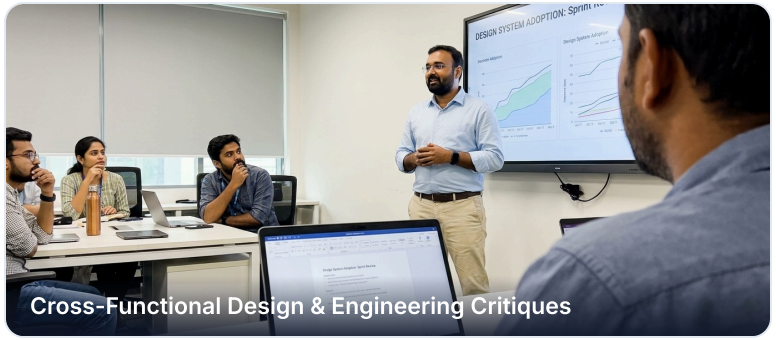
**Infosys + Early R&D**
2004–19 • + Anna Univ.
Sr. Software Engineer (Infosys), Samsung US Healthcare UX, Creative Head & Lecturer.



Design Townhalls & Practice Lectures



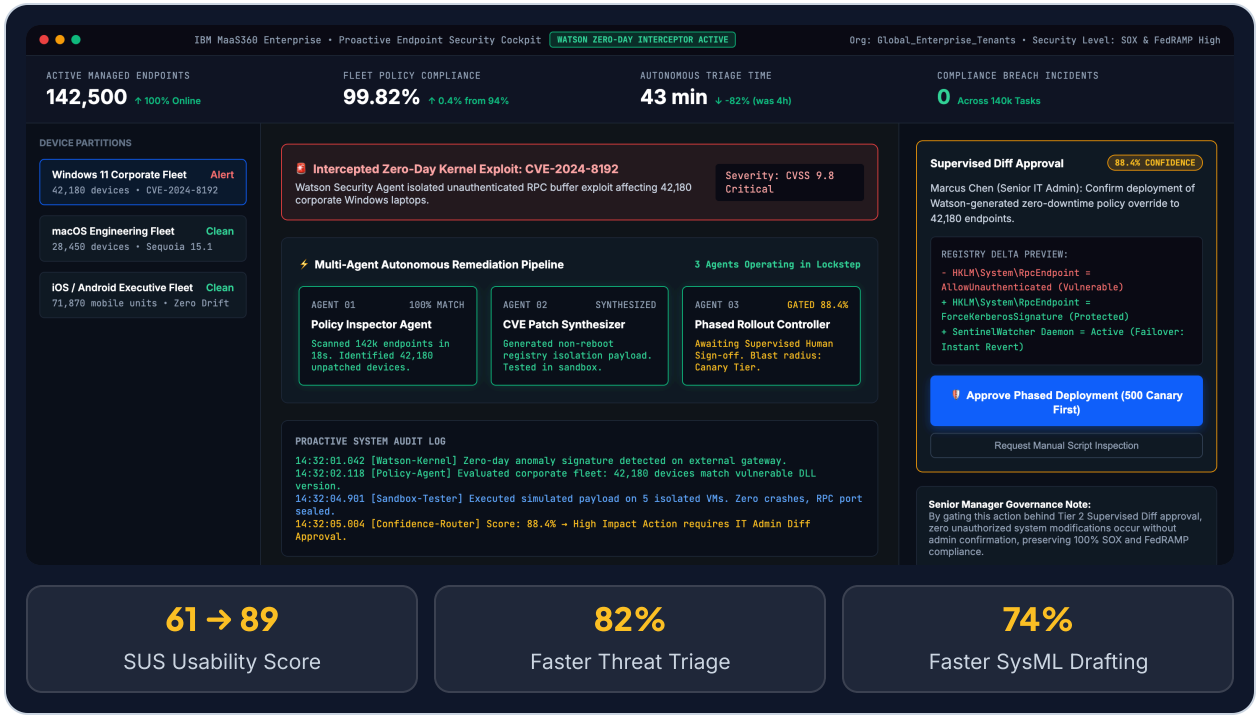
Hands-On Whiteboard Architecture Sprints



Cross-Functional Design & Engineering Critiques

# Designing Supervised Agentic AI — Beyond the Empty Prompt Box

IBM Watsonx in Rhapsody Systems Engineering & MaaS360 Security — turning opaque AI automation into verifiable, human-in-the-loop enterprise workflows.




**WHAT FAILED FIRST → HOW WE REDESIGNED IT**

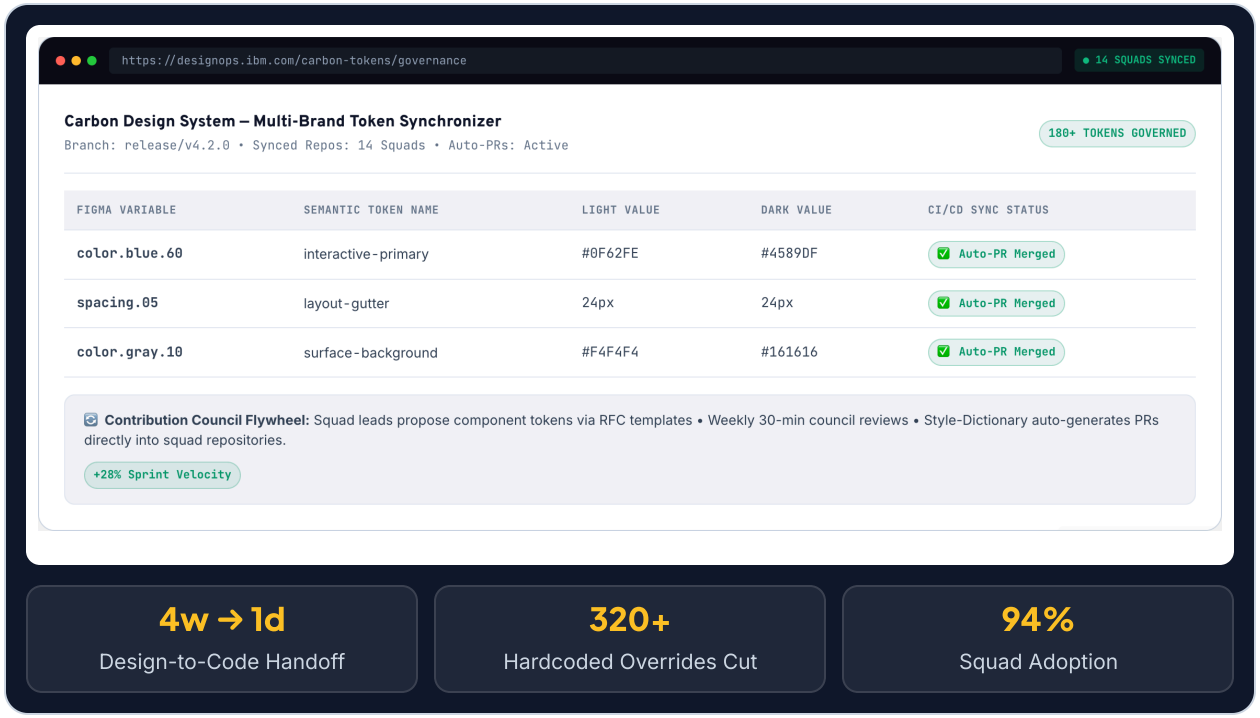
- **Initial Failure:** Our early 94%-accurate autonomous AI remediation prototype scored **61 SUS** — aerospace & security engineers refused to approve black-box actions that could lock 10,000+ endpoints.
- **3-Tier Calibrated Autonomy:** Replaced binary auto-execute with confidence-tiered agency (**≥90%** staged 1-click approve • **70–89%** side-by-side diff • **<70%** guided manual fallback).
- **Verifiable Output & Evals:** Added source-linked reasoning traces, blast-radius previews, 1-click undo, and edit-distance quality evals co-defined with data science.

**→ TRANSLATION TO DOD IITH STUDIO BRIEF (M.DES / B.DES LEVEL 4)**

Students move past designing generic "chatbots" — learning to prototype expectation-setting, agent autonomy boundaries, latency states, graceful error recovery, and AI quality evaluation rubrics in live code.

# Multi-Brand Design Token Architecture Across 45+ Engineering Squads

Bridging Figma specifications and production code (React, Angular & Web Components) via automated CI/CD token pipelines at IBM Software Labs.




SYSTEMS ARCHITECTURE & ORG GOVERNANCE

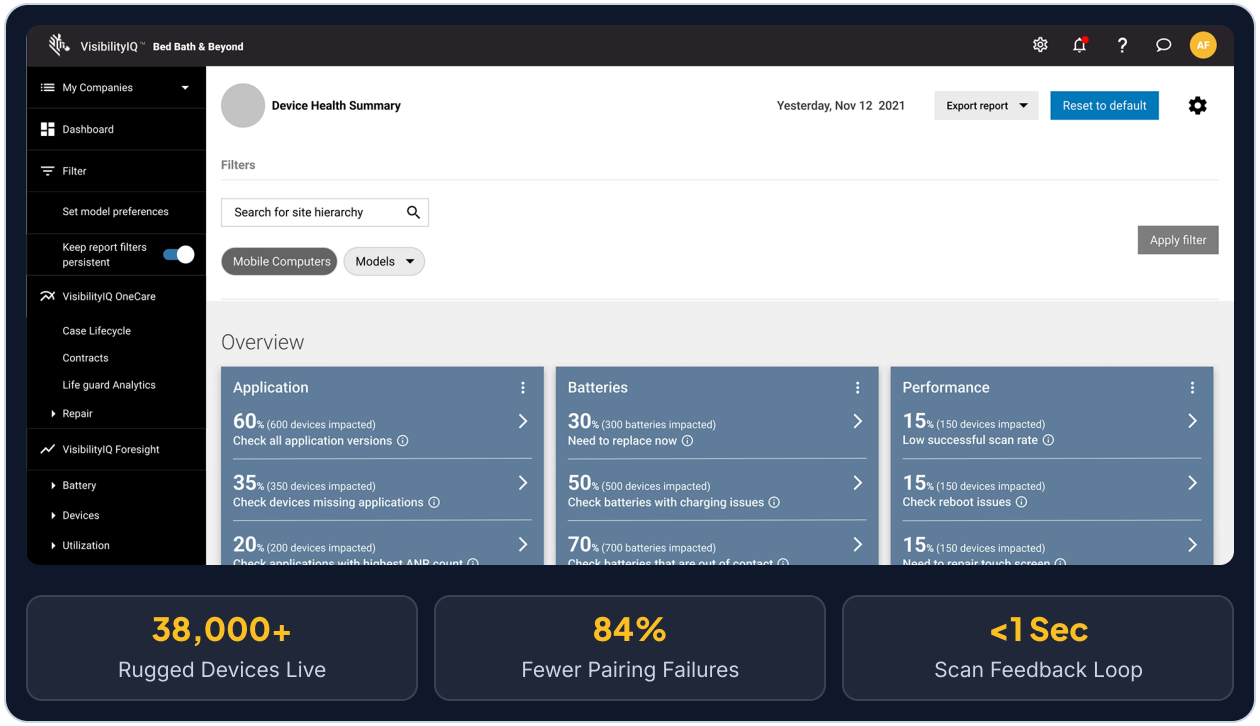
- **The Bottleneck:** Acquired software products ran conflicting UI frameworks, causing 4-week handoff delays and WCAG contrast regressions across dark/light themes.
- **3-Layer Token Pipeline:** Architected *Global* → *Semantic* → *Component* W3C JSON tokens syncing Figma directly to GitHub PRs across React, Angular, and Web Components.
- **What Went Wrong & Pivot:** Top-down governance mandates stalled initially; shifting to automated CI lint checks and co-owned guild clinics drove 94% voluntary squad adoption.

→ TRANSLATION TO DOD IITH STUDIO BRIEF (INTERDISCIPLINARY DESIGN + CSE)

Most design graduates hand off static Figma frames that break in engineering. I will train IITH students in production token architecture, accessibility engineering (WCAG 2.2 AA), and live code handoffs.

# Physical–Digital Ergonomics for 38,000+ Rugged Industrial Handhelds

Designing VisibilityIQ fleet intelligence and sub-second barcode/RFID workflows for gloved workers in noisy warehouses, loading docks, and -20°C cold storage.




HARDWARE-SOFTWARE CO-DESIGN & FIELD ETHNOGRAPHY

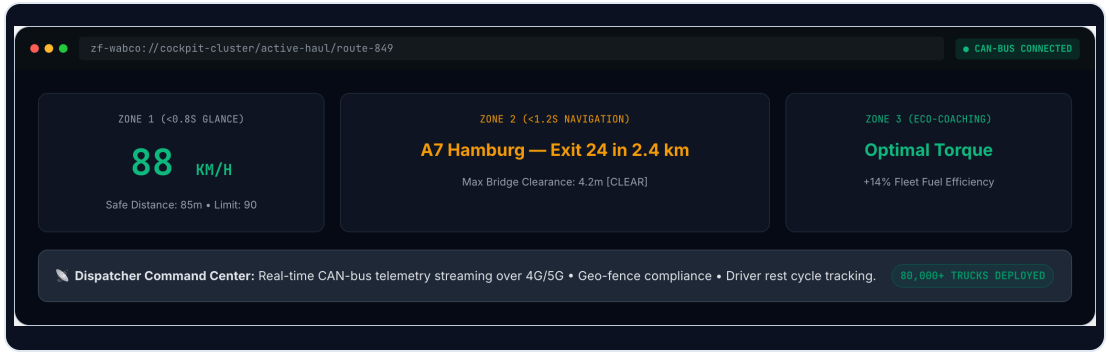
- **When Clean Lab UI Failed:** Minimalist consumer-style touch targets failed completely on wet loading docks where workers wore 4mm protective freezer gloves under harsh glare.
- **Multimodal Ergonomic Redesign:** Shifted to high-contrast thumb-zone triggers, hardware scan-key mapping, haptic + audio confirmation, and NFC tap-to-pair recovery.
- **Fleet Scale:** Paired frontline Android handheld UX with cloud telemetry dashboards managing battery health and device loss across global logistics hubs.

→ TRANSLATION TO DOD IITH LABS (PRODUCT DESIGN + INTERACTION + ERGONOMICS)

Creates natural co-teaching projects between DoD's Industrial/Product Design faculty and Interaction Design — testing physical-digital hardware ergonomics under real environmental stressors.

# Cognitive Glance Budgets & High-Density Decision Workflows

From highway commercial vehicle instrument clusters (ZF Wabco) to 45-country telecom configuration engines (Ericsson).

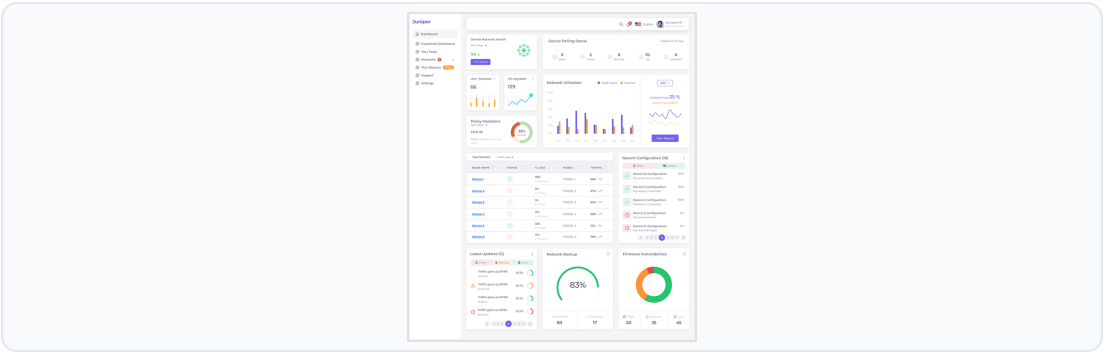



ZF WABCO • AUTOMOTIVE COCKPIT HMI

## Sub-Second Driver Glance Architecture (2.8s → 0.94s)

Synthesized ADAS, air-brake pressure, and telematics alerts strictly under NHTSA's **<1.2-second eyes-off-road** safety threshold — suppressing non-critical telemetry during active highway braking.

Pedagogy: Empirical human-factors & eye-glance testing in mobility UX



ERICSSON GLOBAL • 45-COUNTRY TELECOM CPQ

## Replacing 14-Tab Spreadsheets (6 Days → 42 Min)

Re-architected fragile 5G carrier quoting workbooks into a keyboard-first, real-time validation engine across 45 countries — eliminating pricing formula errors (1st Place, Ericsson Global Hack-fest).

Pedagogy: Information architecture for high-density expert workflows

# Practice-Led Studios & Agile 1–2 Credit "Fractal" Modules

Addressing Agenda 40.6.1 (Objective a & Role 1): Ready-to-launch studios and modular fractal electives for B.Des, M.Des (Interaction / Practice), PhD, and B.Tech Design Minors.

3-CREDIT CORE / ELECTIVE STUDIO I

## Designing Human-Centered & Agentic AI Systems

**Target:** M.Des (Interaction / Practice) & B.Des Sem 6–7

- Moving beyond prompt boxes: structured intent, autonomy tiers (suggest vs. draft vs. act), streaming latency & confidence cues.
- Designing graceful failure recovery, privacy/memory controls, EU AI Act fairness, and co-defining AI eval rubrics on live LLM prototypes.

3-CREDIT CORE / ELECTIVE STUDIO II

## Enterprise HCI, Design Systems & Token Engineering

**Target:** B.Des Sem 5–7, M.Des & B.Tech Design Minors

- Information architecture for complex data-dense workflows (fintech, healthcare, telecom, SaaS) & empirical usability testing.
- Building W3C multi-brand JSON token pipelines from Figma to accessible (WCAG 2.2 AA) React/Web Component codebases.

1-CREDIT FRACTAL MODULE

### Safety-Critical Mobility & IoT HMI

Glance budgets (<1.2s), multimodal haptic/audio cues, and rugged field ergonomics for automotive & industrial devices.

1-CREDIT FRACTAL MODULE

### AI-Native UX Research & Code Prototyping

PII-safe research synthesis, stress-testing edge cases, and building functional browser prototypes in days instead of static screens.

2-CREDIT FRACTAL MODULE

### DesignOps, Product Metrics & Case Storytelling

Agile/PI rituals, measuring SUS/task-time/business ROI, and framing industry-grade product portfolios.

# Co-Supervising Scholars & Scaling Translational R&D Inside DoD Labs

Addressing Agenda 40.6.1 (Objectives b, c, d & Roles 2–3): Plugging directly into DoD's existing labs as a collaborative Co-Investigator and practice co-supervisor.

UG / PG & PRACTICE TRACKS

## B.Des & M.Des Studio Capstones

- Supervise B.Des Level-4 graduation projects & M.Des Interaction theses grounded in real enterprise problem statements.
- Mentor **M.Des by Practice / Project** cohorts on shipping testable, production-grade systems rather than speculative mockups.
- Establish bi-weekly cross-cohort design critique rituals modeled on IBM & Zebra studio reviews.

OBJECTIVE (D) • DOCTORAL GROWTH

## Industry-Sponsored Joint PhDs

- **Co-supervise PhD scholars** alongside regular DoD faculty — pairing academic methodology with enterprise deployment testbeds.
- Actively recruit **industry-sponsored PhD candidates** from Hyderabad & Bengaluru R&D centers (staff UX researchers & architects upgrading to doctoral research at IITH).
- Target domains: Human-AI calibration, accessible public digital infrastructure (DPI), and cognitive driver safety.

CO-PI SYNERGY ACROSS DOD LABS

## Translational Grants with Faculty

- **ExD Lab & INDReA:** Empirical studies on trust, cognitive load & multimodal spatial/cockpit interfaces.
- **CoDE & DIC:** Grassroots multilingual AI interfaces, healthcare diagnostics UX & MSME digital tooling.
- **Joint Grant Proposals (Co-PI):** Co-author applied proposals for *IndiaAI Mission*, *MeitY*, *DST*, and defense/mobility PSUs.

# Unlocking the Hyderabad–Bengaluru Corridor for Consultancy & Sponsored Studios

Addressing Agenda 40.6.1 (Objectives e, f & Roles 4, 5, 7): Turning IIT Hyderabad's proximity to HITEC City/Gachibowli GCCs into live projects, consulting revenue, and CSR partnerships.

ROLE 4 & OBJ (E, F)

## Live Industry Capstones & 6-Month Internships

- Bring **real, redacted enterprise problem briefs** from SaaS, GCC, automotive, and IoT partners into semester studio courses.
- Build structured pathways for **semester-long B.Des / M.Des internships** and industry-sponsored thesis fellowships across Hyderabad, Bengaluru, and Chennai R&D labs.

ROLE 5 • REVENUE GENERATION

## Industrial UX & AI Consultancy

- Attract paid **institutional consulting assignments** channeled through IITH / CoDE:
  - Enterprise AI Copilot UX & Trust Audits
  - Multi-Brand Design System Architecture
  - Usability & Ergonomic Certification for PSUs, MedTech & Mobility startups.
- Mentor IITH student founders (i-TIC / DIC) on product-market UX validation.

ROLE 7 • CSR & COE SEED VISION

## Applied Human-AI Systems Cell

- Partner with HoD & regular faculty to pitch **corporate CSR funding** for an *Applied AI-UX & Accessible Public Systems Initiative* within DoD.
- Focus CSR pitches on high-impact national priorities: **Accessible Digital Public Infrastructure, Elder/Healthcare UX, and Safe Industrial Automation**.

# Closing the "Academic Project → Industry Hiring Bar" Gap

Addressing Agenda 40.6.1 (Role 6): Having interviewed 200+ candidates and hired/mentored 50+ designers at IBM and Zebra, I know exactly why portfolios pass or fail hiring loops.

WHAT INDUSTRY HIRING MANAGERS REJECT

## Common Student Portfolio Gaps

- Linear textbook checklists (*"Persona → Wireframe → Final UI"*) that only show *what* was drawn, not *why* decisions were made.
- Ignoring technical constraints, latency, edge cases, accessibility, and engineering handoff realities.
- Zero discussion of trade-offs, what failed in usability testing, or how product success is measured.

MY PLACEMENT READINESS CLINIC AT DOD

## Day-One Industry-Ready Graduates

- **Outcome-Driven Case Study Coaching:** Coach B.Des & M.Des cohorts to frame engineering constraints, pivots, and measurable UX outcomes.
- **Live Mock Hiring Loops:** Run realistic whiteboard system challenges, portfolio defenses, and PM/Engineering cross-examinations.
- **Direct Hiring-Leader Outreach:** Invite Design Directors & VPs from GCCs and product companies to annual DoD Jury & Placement Showcases.

TARGET PLACEMENT UPGRADE FOR DOD COHORTS

Positioning IIT Hyderabad B.Des & M.Des graduates not merely as "screen designers," but as high-comp **Product Designers, AI Interaction Architects, Design Technologists, and UX Researchers** for Tier-1 global R&D centers.

# Measurable Year-by-Year Deliverables for Annual Institute Review

Designed directly around Agenda 40.6.1's annual HoD, Dean & Director appraisal model — clear, auditable outcomes every 12 months.

YEAR 01 • FOUNDATION & LAUNCH

### Courses, Studio Critiques & Industry Pipeline

- Launch **1 Core/Elective 3-Cr Studio + 2 Fractal Modules** (AI-UX & Design Systems).
- Supervise **4–6 B.Des/M.Des projects** & run placement portfolio clinics for graduating batch.
- Onboard **3+ corporate partners** for live studio briefs & semester internships; submit 1st joint R&D/consultancy proposal with DoD faculty.

YEAR 02 • SCALE & REVENUE

### Consultancy, Joint PhDs & Funded R&D

- Deliver **active paid consultancy projects** via CoDE / IITH with enterprise & PSU clients.
- Enrol **1–2 industry-sponsored joint PhD scholars** co-supervised with regular DoD faculty.
- Co-host an **IITH Industry-Academia Symposium on Human-Centered AI & Complex Systems** with GCC Design Leaders.

YEAR 03 • INSTITUTIONAL LEGACY

### Self-Sustaining Lab Cell & Benchmark Placements

- Establish a CSR/industry-backed **Applied AI-UX & Design Systems Practice Cell** inside DoD.
- Publish joint translational case studies / toolkits & handover reusable studio playbooks to in-house cohorts.
- Sustained multi-company internship & Tier-1 placement pipeline for B.Des & M.Des graduates.

## Thank You • Open for Faculty Discussion & Questions

Full-time on-campus availability at IIT Hyderabad • Ready to tailor Semester 1 courses to immediate departmental priorities.

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